

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Architected a **distributed executor** exposing typed **REST APIs** over Docker and serverless compute with an **S3** store at **sub-200ms p99**, secured with **OAuth/RBAC**.
- Led a production **multi-agent platform** on **AWS** handling **millions of daily transactions** at **99%+ uptime**, and led its architecture and design reviews.
- Shipped **SSE-based streaming** agent delivery with **guardrails** and real-time **observability**, cutting redundant API calls **90%**.
- Built **evaluation harnesses** measuring LLM output quality, safety, **hallucination rates**, and grounding across agent versions, gating releases with **regression suites**.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Cut **p99 latency from 420ms to 140ms** across **50K+ daily requests** by moving LLM serving to an **async, autoscaled** architecture.
- Built **PySpark** jobs and **Airflow** DAGs for large-scale feature and dataset preparation feeding training and **evals**.
- Raised **top-3 retrieval recall from 61% to 89%** on a 2K-query eval set by rebuilding the pipeline with **hybrid dense-sparse** retrieval and re-ranking.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Architected and operated **25+ production Python and Java microservices** for **1M+ users** in **financial services** at **99.9% uptime** on **AWS**.
- Designed high-throughput **SQL and NoSQL** data pipelines on **AWS SQS**, cutting batch processing from **8 minutes to 90 seconds** through **partitioning**, indexed views, and removing lock contention.
- Built **Java**, **Python**, and **C++** services, standardizing retry, timeout, and failure-propagation contracts to cut escalations **60%**.
- Led **architecture and code reviews** and mentored engineers on system design, reliability, and secure API practices.

PROJECTS

Distributed KV Store

github.com/ganeshhgupta/distributed-kv-store

- **Raft-consensus** key-value store with **linearizable reads** and **log compaction**, plus a companion VPC control plane simulator.

LangFetch

tinyurl.com/LangFetch

- **LangGraph** agent that decomposes natural-language **financial queries** into dependency-ordered **SQL** subqueries with **Pinecone** retrieval, served via **FastAPI** as an **MCP** server.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- **LangGraph** multi-agent system that routes production alerts to specialized sub-agents via **tool-use** chains and writes structured root-cause reports, cutting **MTTR 65%**.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Aug 2016 – Jul 2020

SKILLS

Languages: Java, C++, Python, Go, SQL

Distributed Systems: Raft Consensus, Replication, Partitioning, Linearizability, Log Compaction, Stream Processing, Kafka, SQS

Storage & AWS: DynamoDB, PostgreSQL, Redis, S3, Lambda, EC2, Indexing and Query Optimization

Practices: Object-Oriented Design, Design Reviews, Code Reviews, CI/CD, Operational Excellence

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer